BENCHMARK DOCUMENTATION

# Metrics reference

Every quantity reported by the colored point-cloud completion benchmark: how it is computed, what it can establish, and — just as important — what it cannot. Written so a number in a W&B table can be read without opening the notebook that produced it.

`Colored_Point_Cloud_Completion` `arms: RePaint-part · joint-6D (PDR)` `3 categories × 3 difficulties`

## 1 · What is measured, and where

Every model contributes a ground-truth cloud X of 8192 colored points, split by the occlusion protocol into a **visible** part V (the input) and a **missing** part M. The split is a genuine partition — $V \cup M = X$ exactly — so the target is labelled ground truth, not an approximation.

**All color error is measured on M only.** This is not a detail. The visible region is handed to the method as input; including it in the average lets a method score well by copying what it was given. On `simple` that would dilute the measurement by a factor of four.

THE CORRESPONDENCE PROBLEM

A method that also generates geometry does not produce a color *at* the ground-truth points — it produces its own cloud G somewhere near them. To score it, each missing ground-truth point $x \in M$ is matched to its nearest generated point $i(x) = \mathrm{argmin}_{g \in G} \lVert x - g \rVert$, and that point's color is the prediction.

This makes the color score depend on geometric accuracy. If $i(x)$ sits 10 % of the object's radius away, the color fetched is the color of somewhere else. Section 7 exists to remove this dependency.

## 2 · Color error: ΔE in CIELAB

RGB distance is not perceptual — the same numeric step is a large visible change in some regions of the cube and invisible in others. Colors are therefore converted to CIELAB, a space built so that Euclidean distance approximates perceived difference, and compared there.

```
sRGB → linear:   C_lin = C/12.92                    if C ≤ 0.04045
                 = ((C + 0.055)/1.055)^2.4     otherwise

linear → XYZ:    [X Y Z]ᵀ = M · [R G B]_lin      (sRGB primaries, D65)

XYZ → Lab:       t = X/Xn, Y/Yn, Z/Zn           Wn = (0.95047, 1.0, 1.08883)
                 f(t) = t^(1/3)                  if t > δ³,  δ = 6/29
                      = t/(3δ²) + 4/29           otherwise
                 L* = 116·f(Y/Yn) − 16
                 a* = 500·(f(X/Xn) − f(Y/Yn))
                 b* = 200·(f(Y/Yn) − f(Z/Zn))

ΔE (CIE76)  =  √( ΔL*² + Δa*² + Δb*² )
```

The reported number is the **mean ΔE over the missing points**. The scale is absolute and carries a physical interpretation:

| < 1 | 1 – 2 | 2 – 10 | 10 – 30 | > 30 |
|---|---|---|---|---|
| Not perceptible to the human eye | Perceptible on close inspection | Perceptible at a glance | Related colors, clearly different | Effectively unrelated colors |

Commonly cited bands; the classic just-noticeable difference under controlled viewing is $\Delta E \approx 2.3$.

KNOWN LIMITATION

CIE76 is not perceptually uniform everywhere — it overstates differences in saturated blues and understates them in some neutrals. CIEDE2000 corrects this but is not a metric (it violates the triangle inequality), which makes averages and comparisons harder to defend. We report CIE76 for interpretability and state the choice; a paper should say the same rather than imply uniformity it does not have.

## 3 · Why ΔE structurally favours averaging

ΔE is a **distortion** metric. Like any expected-error measure, it is minimized by the conditional mean of the target given the observation, not by a plausible sample from the posterior:

$$\operatorname{argmin}_{\hat{y}} \; E[\, \|\hat{y} - y\|^2 \mid \text{observation} \,] \;=\; E[\, y \mid \text{observation} \,]$$

The consequence is structural, not incidental. A method that predicts the average color of what it cannot see will beat a method that commits to one specific, sharp, plausible answer — *even when the second output looks obviously better to a human*. Blau & Michaeli (2018) prove distortion and perceptual quality are in direct tension: improving one past a point must cost the other.

Two of our baselines are pure averaging procedures, and are therefore advantaged by construction:

| BASELINE | PROCEDURE | WHY IT IS HARD TO BEAT ON ΔE |
|---|---|---|
| `NN-copy` | Each predicted point takes the color of its nearest *visible* point | Locally optimal under spatial color smoothness; commits to no new information |
| `part-mean` | Each point takes the mean color of its part | Literally the conditional mean, given oracle part labels |
| `global-mean` | Every point takes the mean color of the visible region | The coarsest conditional mean; a floor, not a competitor |

HOW TO REPORT RESPONSIBLY

A generative method losing on ΔE is *expected* and is not by itself evidence that it is worse. The honest framing pairs ΔE with a distribution-level measure, and reports the posterior mean of the generative method explicitly (Section 7) so the sampling penalty is separated from model quality.

## 4 · Geometry error: Chamfer distance

$$\begin{aligned} CD(A, B) = {} & (1/|A|) \textstyle\sum_{a\in A} \min_{b\in B} \|a - b\|_2 \\ & + (1/|B|) \textstyle\sum_{b\in B} \min_{a\in A} \|a - b\|_2 \end{aligned}$$

Symmetric, and L1 in the sense that distances enter unsquared — squared Chamfer lets a handful of outliers dominate. Both terms matter and mean different things: the first penalizes predicted points that sit nowhere near the surface (*precision*), the second penalizes surface left uncovered (*recall*).

Clouds are centred and divided by their maximum radius before scoring, so the object fits a unit ball and Chamfer reads as a fraction of object size: **CD = 0.05 means roughly 5 % of the object's radius**. This makes values comparable across categories.

#### The ratio to doing nothing

```
chamfer_vs_partial_x  =  CD(partial, GT) / CD(generated, GT)
```

Raw Chamfer answers "how good", this answers "was it worth running the model". Above 1, completion beats handing back the input untouched; at 1 the model has added nothing; below 1 it has made things worse. The denominator flips it so higher is better, matching every other "×" column.

> READ THIS ONE CAREFULLY
>
> The partial cloud is a *subset of the ground truth*: every point it contains is exactly right. Its Chamfer is non-zero only through the recall term. So the ratio compares a high-precision / low-coverage cloud against a full-coverage / lower-precision one, and it is expected to sit near 1 when little is occluded. That is a property of the comparison, not a failure of the model.

## 5 · Difficulty

Occlusion follows the PoinTr / ShapeNet-55 protocol: draw a random unit viewpoint `v`, sort all points by distance to `v`, and remove the nearest `r·N`. The seed is recorded, so every partial is reproducible from the ground truth alone.

| LEVEL | CROP RATIO R | VISIBLE | WHAT IT PROBES |
|---|---|---|---|
| `simple` | 0.25 | 75 % | Refinement — most of the object is given |
| `moderate` | 0.50 | 50 % | Balanced completion |
| `hard` | 0.75 | 25 % | Genuine generation from a fragment |

> EMPIRICAL FINDING — THIS AXIS DOES NOT CONTROL COLOR DIFFICULTY
>
> Across a category, moving from 25 % to 75 % occlusion roughly doubles Chamfer but moves ΔE by only about 2 units. Across categories, ΔE varies threefold (chair 12–18, airplane 21–27, car 30–38). Color difficulty is set by **texture complexity**, not by how much is hidden. A colored-completion benchmark that inherits difficulty levels from a geometry benchmark is therefore mislabelling its own axis — a second, color-specific axis is needed.

## 6 · Metric catalogue — main run

Logged per model to `joint6d/<category>/<difficulty>`, aggregated into `SUMMARY-joint6d`.

| COLUMN | BETTER | DEFINITION |
|---|---|---|
| `chamfer_gen` | lower | CD(generated, GT) — geometric error of the produced cloud |
| `chamfer_partial` | – | CD(partial, GT) — reference, the cost of returning the input unchanged |
| `chamfer_vs_partial_x` | higher | Ratio of the two; > 1 means completion earned its keep |
| `dE_joint6d` | lower | Mean ΔE on `M`, prediction fetched through the nearest-generated-point map `i(x)` |
| `dE_nn_on_gen` | lower | Same map, but colors come from NN-copy instead of the model. **Same geometry, same correspondence** — the difference isolates coloring |

Pairing `dE_joint6d` with `dE_nn_on_gen` rather than an absolute threshold is deliberate: both travel through the identical geometry and the identical nearest-neighbour map, so whatever error those introduce is common to both and cancels in the comparison.

## 7 · Metric catalogue — ablation

The main-run gap conflates three separate effects. The ablation notebook trains nothing; it reloads the checkpoints and separates them.

### free — the main-run setup, with K samples

Geometry and color both generated. `K = 5` samples per model.

### anchor — geometry pinned to ground truth

At every diffusion step the xyz channel is replaced by the correctly-noised ground-truth geometry, so only color evolves freely — the channel-wise form of RePaint's replacement trick:

```
for t = T−1 … 0:
    x  ←  [ q_sample(X_gt, t) ‖ x_rgb ]      # xyz pinned at noise level t
    x  ←  p_sample( model(x, t, cond), x, t )
```

Generated color then lands *exactly on* the ground-truth points, `i(x)` becomes the identity, and the correspondence drops out of the measurement entirely. What remains is pure color error. The model was not trained this way — it is an inference-time diagnostic, not a claim about the trained model.

| COLUMN | DEFINITION |
|---|---|
| `dE_free_sample` | Mean over K samples of the per-sample ΔE. Reproduces the main-run metric |
| `dE_free_postmean` | ΔE of the *color-space average* of the K samples — the posterior-mean estimate ΔE actually rewards |
| `dE_free_nn` | NN-copy on the generated geometry (main-run baseline, recomputed as a control) |
| `dE_anchor_sample` | Per-sample ΔE with geometry pinned to GT |
| `dE_anchor_postmean` | K-sample color average, geometry pinned to GT — **the cleanest measure of the model's coloring** |
| `dE_anchor_nn` | NN-copy evaluated on GT geometry — the baseline on identical footing |
| `dE_anchor_global` | Every missing point painted the mean color of the visible region — the prior floor |

## 8 · Derived quantities, and how to read them

| QUANTITY | DEFINITION | READING |
|---|---|---|
| `sampling_penalty` | `free_sample − free_postmean` | **Positive** → part of the gap is the variance cost of drawing one sample, not model quality. Large values mean ΔE is penalizing the act of sampling |
| `geometry_penalty` | `free_sample − anchor_sample` | **Positive** → the nearest-neighbour correspondence was inflating the color score. This is measurement error, not coloring error |
| `gap_free` | `free_sample − free_nn` | The main-run headline, recomputed. Serves as a consistency check against the original run |
| `gap_anchored` | `anchor_postmean − anchor_nn` | **The decisive number.** Both sides on ground-truth geometry, both at their best form. Negative means the generative model genuinely beats NN-copy |

| QUANTITY | DEFINITION | READING |
|---|---|---|
| `prior_margin` | `anchor_global − anchor_postmean` | **Positive** → the model beats a constant color, so conditioning is doing something. **Near zero** → the color head is emitting a category prior and ignoring the visible colors |

THE DECOMPOSITION

Read together: `gap_free` is what was published; `sampling_penalty` and `geometry_penalty` say how much of it was method artefact; `gap_anchored` is what survives once both are removed. If `gap_anchored` stays positive, the negative result is real and reportable. If it goes negative, the original conclusion was an artefact of single-sample evaluation through a noisy correspondence — a materially different paper.

## 9 · Reference values

Numbers to anchor any new result against. All are ΔE unless noted.

| QUANTITY | VALUE | WHAT IT ESTABLISHES |
|---|---:|---|
| Within-part ΔE, real texture | 20.7 | Structural ceiling for any part-constant coloring on real ShapeNet texture — no such method can score below this |
| Within-part ΔE, synthetic texture | 5.5 | The same ceiling on flat synthetic color; the 4× gap is why Stage-1 real texture was necessary |
| Textured fraction of models | 74 % | Share of models with genuine texture (4 % flat) — the benchmark is not dominated by trivial cases |
| Joint-6D vs NN-copy, 9 cells | +4.06 ± 0.94 | Systematic, low-variance loss across every category and difficulty |
| RePaint-part vs NN-copy, airplane/hard | +9.61 ± 3.04 | The earlier arm; joint-6D roughly halves both the gap and its spread |
| Chamfer floor, 2048 points | ≈ 0.05 | Best geometric precision observed at this resolution, independent of difficulty |

## 10 · What these metrics cannot tell you

- **Whether a result looks right.** ΔE is a per-point average and is blind to spatial structure — a prediction with correct colors in wrong places can score identically to one with the arrangement right. No distribution- or structure-level metric is currently reported.
- **Whether the method is better for a human viewer.** Distortion and perceptual quality are provably in tension (Section 3). Winning on ΔE is not winning on appearance, and losing on it is not losing on appearance.
- **Anything about categories outside the three benchmarked.** ΔE level is governed by texture complexity, which varies threefold within these three alone. Extrapolation is unjustified.
- **Statistical significance.** Aggregates are means over 15 test models per cell. Spreads are reported where they exist; no significance test is run, and none should be implied.